

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

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1 Business Understanding

This project aims to use smartphone sensor data to classify bicycle-lane sections as smooth or bumpy. Cracks, small potholes, and uneven surfaces can affect cycling comfort and safety. Detecting these sections could help city planners decide where repairs are needed. Smartphone sensors provide a simple way to collect road-condition data during everyday rides and could reduce the need for manual inspections.

The vibration data may also be useful for health-monitoring research, such as studying cycling comfort and the effects of repeated shocks on the body. However, this project only identifies road conditions and does not measure health effects.

The task is a binary classification problem using accelerometer, gyroscope, and gravity sensor data. Each recording is divided into short time windows, and statistical features are calculated for each window. Four supervised models and four unsupervised clustering methods are compared to find useful features and distinguish smooth from bumpy conditions.

The final model will be chosen based on its performance, suitability for the data, ease of understanding, and computing needs. It will then be applied to a separate external dataset to show which sections are predicted as smooth or bumpy. This dataset will not be used for training, validation, or model selection. The goal is to build a working prototype that others can run and explain its limitations.

2 Data Understanding

2.1 Data Collection Design

Three group members participated in the study. Each participant recorded cycling data on a representative smooth lane and a noticeably uneven but safe lane. The Sensor Logger application recorded the accelerometer, gyroscope, and gravity sensor at a target sampling frequency of 100 Hz with the standardization setting enabled. Other sensors were disabled. The smartphone was secured vertically in the bicycle basket throughout each selected ride to keep its placement as consistent as possible.

A recording started before the phone was placed in the basket and ended after the bicycle stopped. This procedure creates short stationary or handling periods at the beginning and end of a file, which must be identified during preprocessing. Smooth routes were chosen from regular paved bicycle lanes without obvious defects. Bumpy routes contained common imperfections such as cracks, uneven paving, or small potholes, while dangerous road conditions were avoided. Participant identifiers P1 to P3 were used in the files and analysis.

Table 1: Selected recordings by participant and lane condition

Participant ID	Phone model	Route	Class	Duration(Sec)	Sample count	Target sampling frequency
P1	Iphone 15	Near Geldropsedijk bus stop	smooth	332.2	33176	100 Hz
P2	Iphone 13	Near Geldropsedijk bus stop	smooth	501.8	49779	100 Hz
P3	Iphone 17 pro max	Circular bicycle path in TU/e	smooth	561.1	55894	100 Hz
P1	Iphone 15	Eisenhowerlaan	bumpy	508.1	50749	100 Hz
P2	Iphone 13	Opwettenseweg	bumpy	510.2	50613	100 Hz
P3	Iphone 17 pro max	Brick-paved path near Luna, TU/e	bumpy	579.3	57708	100 Hz

2.2 Initial Data Quality Checks

The three sensor files from each ride are loaded separately and renamed with sensor-specific column names. Before merging, the pipeline checks whether the accelerometer, gyroscope, and gravity files contain the same number of observations and whether their timestamps are aligned. The merged recording is then checked for missing values, duplicate timestamps, recording duration, and effective sampling frequency. These checks prevent corrupted or misaligned measurements from entering feature extraction.

Table 2: Data-quality and sampling checks

Ride	Missing values	Duplicate timestamps	Effective rate (Hz)
P1 Smooth	0	0	99.9
P1 Bumpy	0	0	99.9
P2 Smooth	0	0	99.2
P2 Bumpy	0	0	99.2
P3 Smooth	0	0	99.6
P3 Bumpy	0	0	99.6

2.3 Exploratory Data Analysis

Accelerometer and gyroscope magnitude plots were used to compare smooth and bumpy recordings. The plots were examined for differences between participants and possible phone-handling effects.

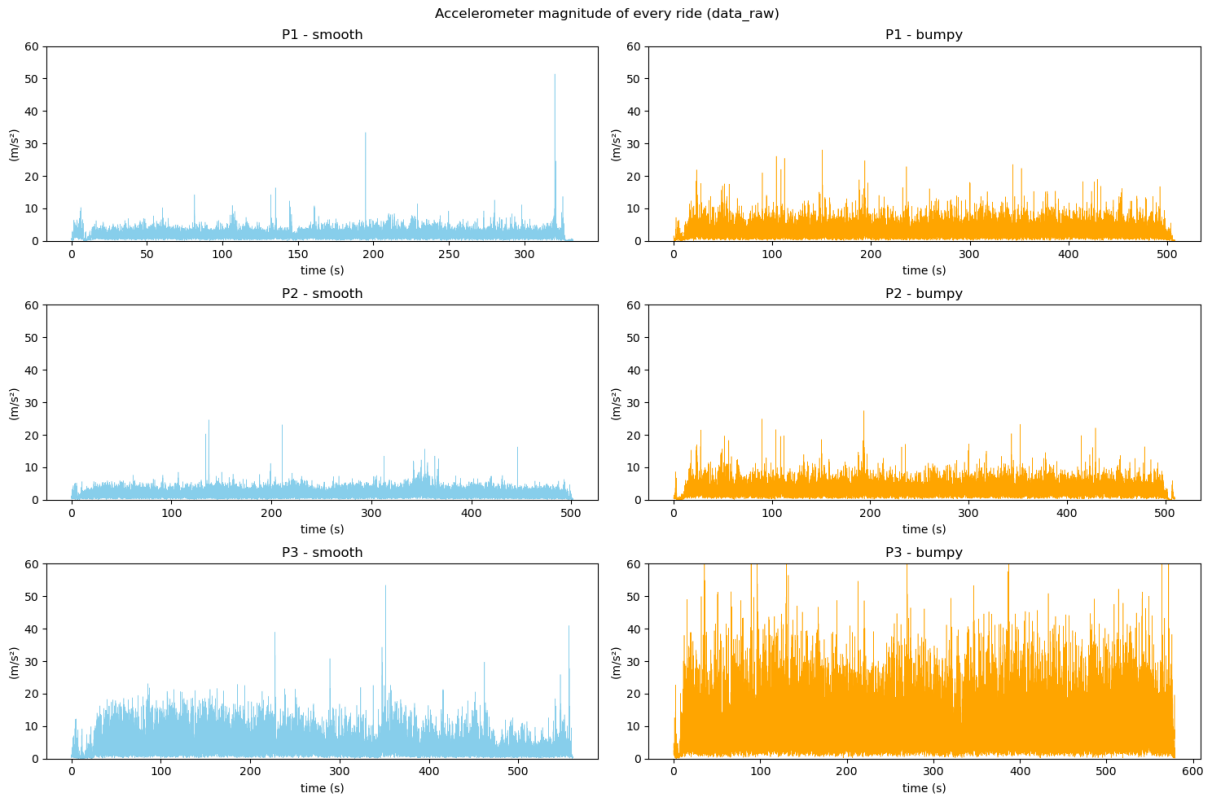


Figure 1: Final accelerometer smooth and bumpy sensor comparison.

The P1 and P2 bumpy rides, shows a thicker, more variable band with a frequent spikes, while smooth rides are mostly lower with few spikes. For P3 its smooth remains elevated after 30 seconds, and in bumpy ride it recorded maximum of 88.72 m/s², the cut-off line is at 60 m/s². The figure provides the information about untrimmed signals, quiet start-up ride remain labeled according to ride condition, which helps explain the difference between participants.

The gyroscope magnitude plots are provided in the Appendix (Figure 6). The smooth recordings of P1 and P2 showed a low baseline with sharp peaks near the beginning and end, which may reflect phone handling. P3 showed a higher baseline, possibly due to differences in rider movement, bicycle setup, or route. Gyroscope signals capture

turning and steering as well as rotation caused by bumps.

3 Data Preparation

3.1 Preprocessing

The accelerometer, gyroscope, and gravity were merged using their timestamps. Sensor magnitudes are calculated as the Euclidean norm of the x , y , and z axes. Row counts and timestamps were validated across the three streams. No missing values or duplicate timestamps were detected. There is no resampling, filtering, or trimming of the start and end of the rides. The magnitude was calculated as $\sqrt{x^2 + y^2 + z^2}$, and its standard deviation and maximum has been computed for each participant ride.

Table 3: Ride-level accelerometer-magnitude statistics

Statistic	P1 Smooth	P1 Bumpy	P2 Smooth	P2 Bumpy	P3 Smooth	P3 Bumpy
Standard deviation (m/s ²)	1.33	1.92	1.11	1.90	2.47	5.59
Maximum (m/s ²)	51.38	28.03	24.60	27.42	53.40	88.72

3.2 Windowing and Feature Extraction

The continuous recordings are divided into two-second windows, corresponding to approximately 200 observations at 100 Hz. Consecutive windows begin one second apart, giving a 50% overlap. This length captures short vibration patterns while still producing enough windows for model training. For each window, the pipeline calculates the mean, standard deviation, root mean square, minimum, and maximum of the accelerometer, gyroscope, and gravity magnitudes. The initial feature matrix therefore contains 15 time-domain features.

Gravity-magnitude features showed little variation in the current analysis because the magnitude remains close to gravitational acceleration. These near-constant features provide limited information for separating smooth and bumpy surfaces, so the current pipeline removes them before modeling. The remaining candidate features describe acceleration intensity and rotational movement. Mutual information and correlation analysis are used to examine relevance and redundancy.

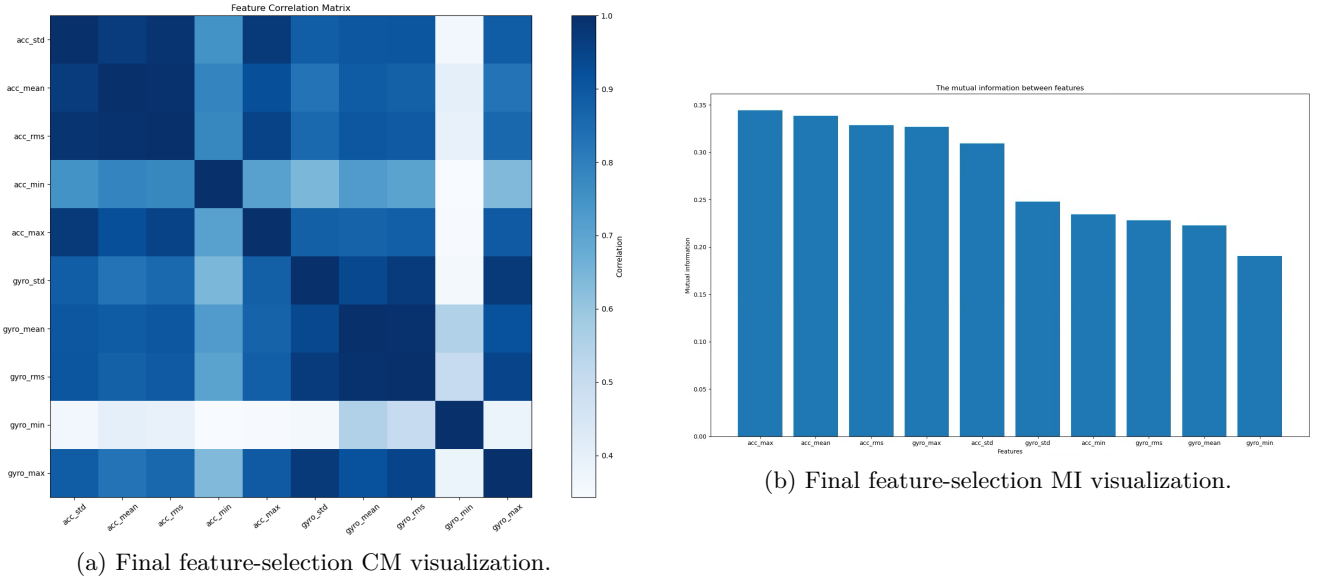


Figure 2: Feature selection analysis

Mutual information ranked `acc_max`, `acc_mean`, `acc_rms`, and `gyro_max` as the most informative individual features. The correlation matrix showed that several features contain similar information. All ten acceleration and gyroscope features were retained for modeling. PCA was then used to reduce the number of inputs while keeping 95% of the variance.

4 Modeling

All models used the ten acceleration and gyroscope magnitude features described in Section 3. StandardScaler was used to put the features on a similar scale. PCA was then applied to reduce the number of inputs while keeping 95% of the variance. The scaler and PCA were fitted on the training data and then applied to the test data.

4.1 Unsupervised Approaches

Four clustering methods were used to explore whether the features form groups that match smooth and bumpy conditions. The models did not use the class labels during fitting. Two clusters or components were chosen because the task has two classes. However, the groups found by the models do not necessarily represent road conditions.

K-means groups windows around cluster centers. Agglomerative Clustering builds groups by merging similar observations. Fuzzy C-Means allows each window to belong partly to both clusters. The cluster with the highest membership was used as its final assignment. Gaussian Mixture Model (GMM) uses two probability distributions to describe the groups in the data.

Table 4: Unsupervised models and settings.

Model	Settings
Agglomerative Clustering	Two clusters; Ward linkage
K-means	Two clusters; random seed 42
Fuzzy C-Means	Two clusters; $m = 2$; tolerance 0.005; maximum 1,000 iterations
GMM	Two components; full covariance; random seed 42

The unsupervised section used a 75/25 random split with seed 42. K-means, Fuzzy C-Means, and GMM were fitted on the training data and predicted clusters for the test data. Agglomerative Clustering was fitted directly on the test data because the implementation does not provide a standard method for predicting new windows. Its results therefore describe clustering within that test set rather than predictions from a training model.

4.2 Supervised Approaches

Four supervised models were trained using the smooth and bumpy labels. These models were included to compare different ways of separating the two classes.

Logistic Regression provides a simple linear baseline. k-Nearest Neighbors (k-NN) predicts the class using nearby training examples. Random Forest combines decision trees and can learn more complex relationships between features. The Support Vector Classifier (SVC) uses an RBF kernel to learn a nonlinear boundary between the classes.

Table 5: Supervised models and settings.

Model	Settings
Logistic Regression	Default settings, including $C = 1$
k-NN	Five neighbors; uniform weights; Euclidean distance
Random Forest	50 trees; maximum depth 10; entropy criterion; random seed 42
SVC	RBF kernel; $C = 1$; gamma set to <code>scale</code>

All supervised models used the same 75/25 random split with seed 42. This gave 2,227 training windows and 743 test windows. The same scaling and PCA steps were used for each model. k-NN used five neighbors to make predictions based on several nearby training examples. Random Forest used 50 trees with a maximum depth of 10 to keep the model size manageable and limit tree complexity. Logistic Regression and SVC used their default settings, with an RBF kernel for SVC. The settings used for each model are summarized in Table 5.

4.3 Comparison Methodology

The supervised models were compared using accuracy, F1 score, recall, and confusion matrices. Bumpy was the positive class for F1 and recall. These measures show how well the models classify windows and how often they detect or miss bumpy sections.

The unsupervised models were compared using Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and silhouette score. ARI and NMI measure how closely the clusters match the known smooth and bumpy labels. They do not require cluster numbers to be matched to class names. Silhouette measures how clearly the clusters are separated without using the labels.

The supervised and unsupervised scores measure different things, so they cannot be compared directly. The approaches were also compared based on their need for labeled data and their usefulness for road-condition prediction. The fitting method used for Agglomerative Clustering was considered when interpreting the results. The final model choice considers performance, suitability for the data, ease of understanding, computing needs, and the ability to predict new windows.

5 Evaluation

This section compares model performance, examines classification errors, and explains the final model choice. The models were evaluated using the random window-level splits described in Section 4: 75/25 for supervised models and 75/25 for unsupervised models. Scaling and PCA were fitted on the training data before transforming the test data.

Because neighboring windows overlap, training and test windows may share sensor samples. The scores may therefore overestimate performance on new rides. The external dataset was reserved for deployment and was not used for model selection.

5.1 Unsupervised Results

Table 6 presents the results of the four clustering methods. ARI and NMI measure agreement with the known road-condition labels, while silhouette measures separation between clusters.

Table 6: Performance of the unsupervised models.

Model	ARI	NMI	Silhouette
Agglomerative Clustering	0.0845	0.1998	0.7039
K-means	0.0861	0.1900	0.7098
Fuzzy C-Means	0.0877	0.1915	0.7099
GMM	0.0880	0.1486	0.6572

GMM achieved the highest ARI, Agglomerative Clustering had the highest NMI, and Fuzzy C-Means had the highest silhouette score. The differences in ARI were small. Overall, all four methods showed low agreement with the smooth and bumpy labels.

Figure 3 compares the true labels with the cluster assignments in PCA space. The true labels overlap within the large main group, while the clustering methods mainly separate this group from a smaller group farther away. This helps explain why the silhouette scores are relatively high, while ARI and NMI remain low: the clusters are separated, but they do not closely follow the road-condition labels.

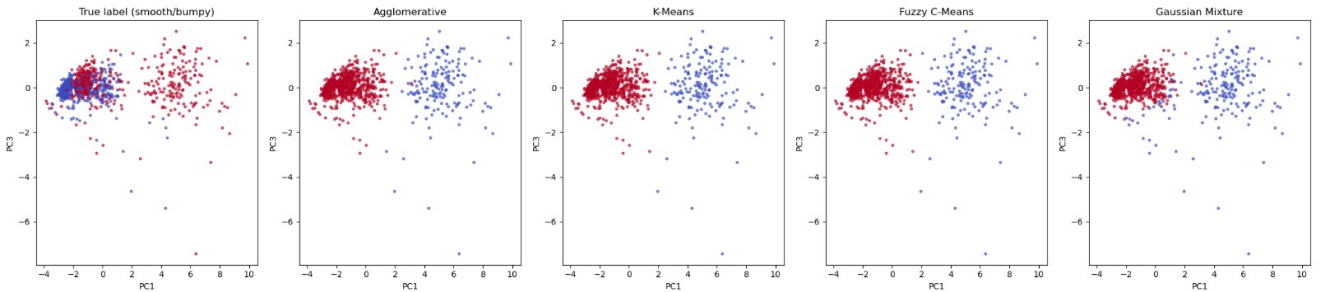


Figure 3: True road-condition labels and cluster assignments shown using the first and third principal components. Blue represents smooth and red represents bumpy; cluster colors follow the label mapping used for comparison.

The groups may reflect differences in vibration strength, riders, or bicycles as well as road conditions. The clustering methods were therefore useful for exploring the data, but provided limited support for identifying smooth and bumpy windows.

Agglomerative Clustering was fitted directly on the test set. Its scores describe clustering within that set, whereas the other methods assigned test windows using models fitted on the training data.

5.2 Supervised Results

Table 7 compares the four supervised models on the same 743 test windows. F1 score and recall are reported for the bumpy class.

Table 7: Performance of the supervised models.

Model	Accuracy	F1 score	Recall
Logistic Regression	0.68	0.68	0.64
k-NN	0.81	0.83	0.85
Random Forest	0.83	0.84	0.88
SVC	0.81	0.83	0.89

Random Forest achieved the highest accuracy and F1 score. SVC achieved the highest recall, meaning that it detected the largest proportion of bumpy windows. k-NN also performed well, while Logistic Regression had the lowest scores. The stronger results of the more flexible models suggest that a simple linear boundary may not fully capture the differences between the two classes.

The differences between Random Forest, k-NN, and SVC were small. Random Forest had the best overall scores on this split, but these results do not establish that it would consistently outperform the other models on different recordings.

Figure 4 shows the confusion matrices of all four models. The test set contained 397 bumpy windows and 346 smooth windows. The diagonal entries represent correct predictions, while the other entries show missed bumpy windows and smooth windows incorrectly predicted as bumpy.

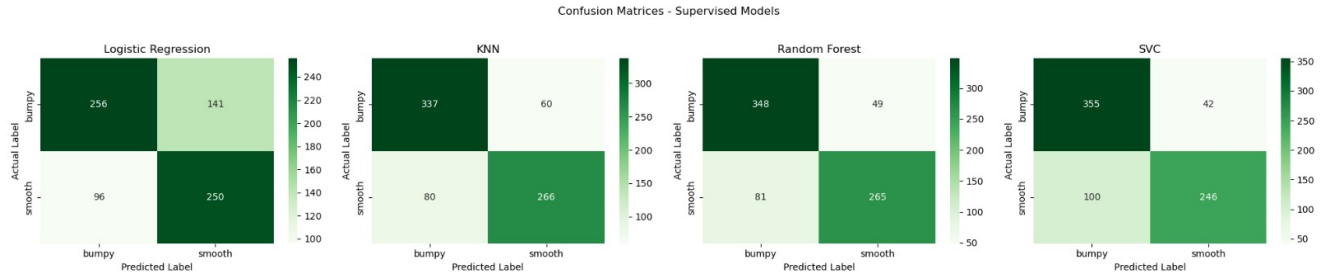


Figure 4: Confusion matrices of the four supervised models on the test set. Rows show actual labels and columns show predicted labels.

5.3 Classification Errors and Failure Modes

The confusion matrices in Figure 4 show two types of classification error. A bumpy window predicted as smooth is a missed detection, which could leave an uneven section unreported. A smooth window predicted as bumpy is a false alarm, which could lead to an unnecessary inspection.

Logistic Regression missed 141 bumpy windows, compared with 60 for k-NN, 49 for Random Forest, and 42 for SVC. Random Forest correctly classified 348 bumpy windows and 265 smooth windows, but produced 81 false alarms. SVC detected seven more bumpy windows than Random Forest, but produced 100 false alarms. Its higher recall therefore came with more smooth windows being flagged as bumpy.

Some errors may occur because smooth and bumpy conditions produce similar measurements. A smooth road can still include short impacts or movement caused by steering, while a bumpy road may produce weaker vibrations at lower speeds. Basket movement may also affect the signals. These are possible explanations rather than confirmed causes of individual errors.

Each recording also received one label for the whole ride. A window may therefore contain movement that is not typical of its assigned class. For example, a ride labeled bumpy may contain a short smooth section. This limits how precisely window-level errors can be interpreted.

5.4 Final Model Selection

Random Forest was selected as the final model because it achieved the highest accuracy (0.83) and F1 score (0.84), together with a high recall (0.88). Compared with SVC, it missed slightly more bumpy windows but produced fewer false alarms. It therefore offered a useful balance for the intended road-assessment task.

Random Forest is suitable for binary classification and can learn complex relationships between the sensor features. It requires labeled training data, which was available for this study, and directly predicts smooth or bumpy classes for new windows. The clustering methods do not require labels during fitting, but their low agreement with the known classes limited their usefulness for this purpose. Their scores measure different aspects of performance and cannot be compared directly with supervised accuracy or F1.

Interpretability and computing needs were also considered. Random Forest is less easy to explain than Logistic Regression, and PCA makes it harder to connect decisions directly to the original sensor features. However, 50 trees with a maximum depth of 10 keep the model size manageable. Prediction speed would still need to be measured before use in a live application.

Based on these considerations, Random Forest was carried forward to the external dataset. The deployment results are presented in Section 6.

6 Deployment Considerations and Recommendations

6.1 Deployment Pipeline and External Results

Random Forest was applied to the independent external dataset. The three sensor files were merged and divided into two-second windows with a one-second step. Statistical features were calculated, followed by scaling, PCA, and model prediction. The external dataset was not used for training, tuning, or testing.

Table 8 summarizes the predictions. Of the 1,018 windows, 638 (62.7%) were classified as smooth and 380 (37.3%) as bumpy. The model therefore predicted more smooth windows than bumpy windows. These percentages describe the share of overlapping windows, not the share of road distance.

Table 8: Random Forest predictions on the external dataset.

Predicted class	Number of windows	Percentage
Smooth	638	62.7%
Bumpy	380	37.3%

Figure 5 shows how the predictions change during the ride. The black line represents the estimated probability of the bumpy class. Red indicates a probability of at least 0.5, while green indicates a probability below 0.5.

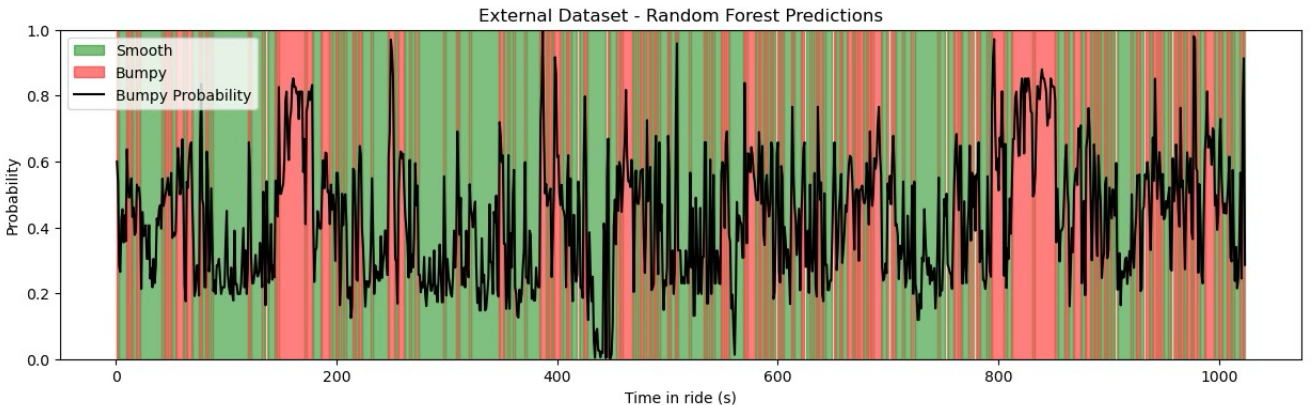


Figure 5: Random Forest predictions over time: green = smooth, red = bumpy, and the black line shows the estimated probability of the bumpy class.

The model predicts sustained bumpy sections at approximately 140–180 seconds and 800–850 seconds, where the probability remains relatively high. In contrast, around 430–450 seconds, the probability falls close to zero, indicating a strong prediction of smooth riding. Elsewhere, predictions frequently switch between smooth and bumpy, with many probabilities near the 0.5 threshold. These windows are less clearly separated by the model.

The timeline helps locate predicted changes in road conditions within the recording. However, no reference labels were used to verify the external predictions, so their accuracy cannot be determined here. Changes in speed, turning, or phone movement may also influence the predictions. The results demonstrate how the model could highlight sections for further inspection.

6.2 Limitations

The study includes only three participants and six selected recordings. This provides a useful starting point, but it does not cover the full range of riders, bicycles, phones, and road surfaces. Each participant recorded one selected ride per class, so the model may learn patterns specific to these routes. Its performance on new participants and routes remains uncertain.

Phone placement is another limitation. Although the phones were secured in bicycle baskets, the baskets themselves may move slightly during cycling. This movement could add vibration to the sensor readings. The measured signals may therefore reflect both road conditions and the movement of the basket. Differences in basket structure and attachment between bicycles could also affect how vibration reaches the phone.

Cycling speed, steering, and rider behavior may influence the measurements. For example, the same uneven surface may produce different vibration levels at different speeds. The gyroscope also records rotation caused by turning and steering, rather than only movement caused by bumps. These effects were not measured separately, so their contribution cannot be clearly distinguished from road roughness.

6.3 Recommendations and Future Work

Future work should collect more rides from different participants, routes, bicycles, and phones. Repeated rides would help check whether the results are consistent. Testing on riders and routes not used for training would show how well the model works on new data.

Phone placement should be consistent. Comparing basket placement with a firm bicycle mount could help show how extra movement affects the readings. GPS and speed data could link predictions to road locations and help explain the effects of stops, turns, and speed changes. Shorter road sections should be labeled separately instead of giving one label to a whole ride. Different window lengths and feature sets should also be tested, including models with and without gyroscope features. Combining predictions from nearby windows could reduce rapid class changes, but this should be checked to avoid hiding short bumps.

In practical use, this tool could help city maintenance teams identify road sections for further inspection. Combining data from several cyclists could provide stronger evidence than a single ride. The predictions could also give cyclists information about route comfort. Vibration data may support research into cycling comfort and health monitoring, but health-related use would require further validation.

6.4 Ethics, Privacy and Reproducibility

Ethics. All participants were adult group members who understood the purpose of the project and agreed to the data collection. Routes were selected to represent normal cycling conditions, and dangerous surfaces were avoided. Participant safety took priority over collecting strong vibration signals.

Privacy. Participants are referred to as P1–P3 in the data analysis and figures. The sensor recordings are used only for this educational assignment. Although these identifiers reduce the use of personal information, phone models and route descriptions are included to explain the collection conditions. The recordings will be deleted after course completion and final grading.

Reproducibility. The notebook contains the code for data loading, feature extraction, model training, evaluation, and external-data prediction. The README provides setup instructions and describes the dataset organization. Python and package versions should be listed so that others can recreate the environment.

7 Appendix

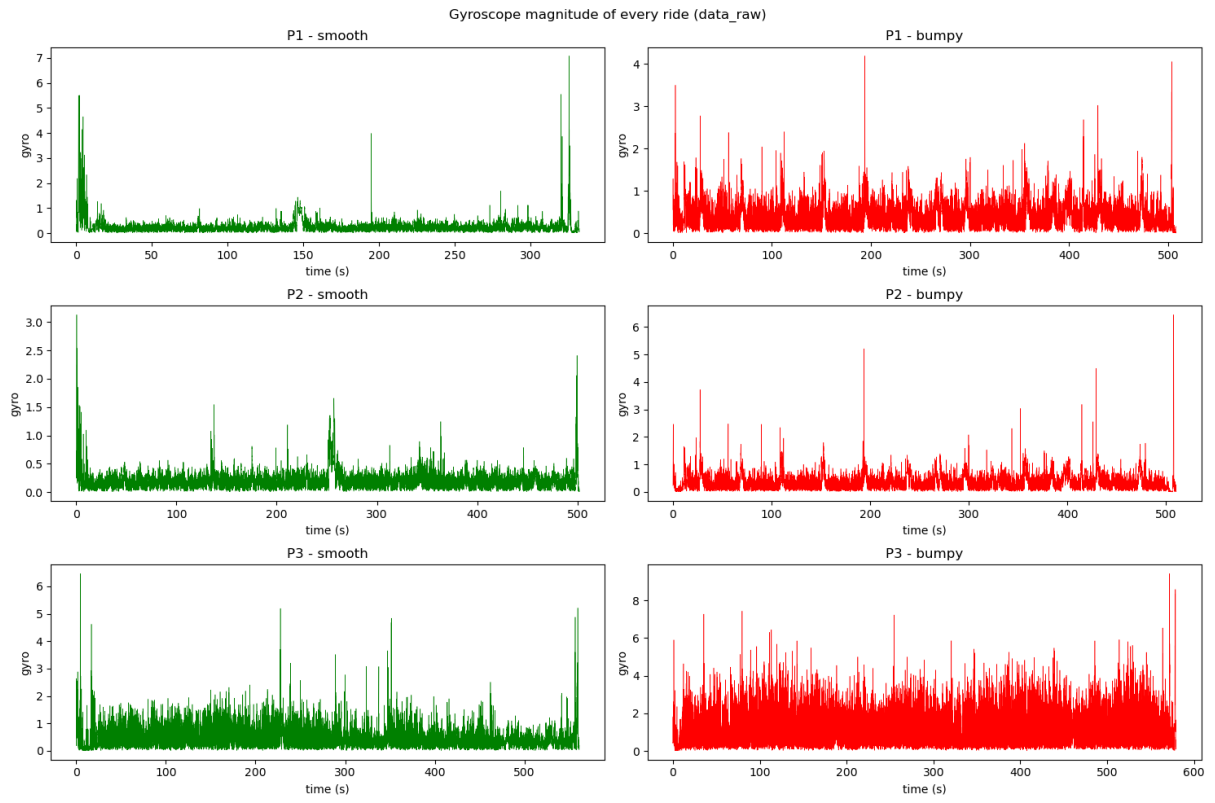


Figure 6: Gyroscope magnitude for the six selected recordings.